INTERNATIONAL SEARCH REPORT

International application No
PCT/US2022/028086

A. CLASSIFICATION OF SUBJECT MATTER
INV. G01J5/35 H05B47/13 H05B47/16 G08B13/191
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G01J H05B G08B H03F

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X A	JP 2010 230492 A (PANASONIC CORP) 14 October 2010 (2010-10-14) paragraphs [0010], [0030], [0048], [0052] - [0059] figure 2	1-28, 38-48 29-37
X	JP H05 81670 U (NN) 5 November 1993 (1993-11-05) paragraphs [0002], [0003] figures 1, 2, 4	24-28
A	JP H04 336806 A (SEKISUI CHEMICAL CO LTD) 25 November 1992 (1992-11-25) paragraphs [0005] - [0010] figures 1-6	1,24,38

-/--

[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
18 August 2022	02/09/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Hillmayr, Heinrich

1

Form PCT/ISA/210 (second sheet) (April 2005)

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2022/028086

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US R E47 511 E (BLAKELEY MATTHEW ROBERT [US] ET AL) 9 July 2019 (2019-07-09) column 10, line 3 - column 11, line 18 column 22, line 34 - line 42 figures 3, 18	1,24,38
A	JP S57 170607 A (OPT KK) 20 October 1982 (1982-10-20) paragraph [0001] - paragraph [0002] figure 1	1,24,38

1

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2022/028086

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
JP 2010230492 A	14-10-2010	NONE	
JP H0581670 U	05-11-1993	NONE	
JP H04336806 A	25-11-1992	NONE	
US RE47511 E	09-07-2019	NONE	
JP S57170607 A	20-10-1982	JP H0334243 B2	22-05-1991
		JP S57170607 A	20-10-1982